

Literature Status: Polynomially Bounded Class- C_{11} Operators

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Status: literature already answered, negatively. Question 18 of Kérchy asks whether a polynomially bounded operator of class C_{11} must be similar to a contraction. Gamal' explicitly states that her construction answers this question negatively: it is polynomially bounded and quasisimilar to a unitary, but is not similar to a contraction.

1 Original question

László Kérchy, *Quasianalytic polynomially bounded operators*, arXiv:1503.06691 [1], asks on arXiv PDF page 18:

Question 18. If the polynomially bounded operator T is of class C_{11} , is T similar to a contraction?

Here C_{11} means that neither T nor T^* has a nonzero stable vector. The source notes immediately before the question that such an operator is quasisimilar to its unitary asymptote.

2 Separate later answer

Maria F. Gamal', *Examples of cyclic polynomially bounded operators that are not similar to contractions*, arXiv:2412.14130 [2], states on arXiv PDF page 2 that the paper provides a negative answer to Question 18 of Kérchy [1]. The same paragraph says that the construction produces polynomially bounded operators which are not similar to contractions and are quasisimilar to the bilateral shift and to cyclic reductive unitaries. It derives these examples through the Section 2 transfer construction and the main Theorem 7.1 (PDF page 24).

For completeness, quasisimilarity to a unitary indeed places the example in the source's class C_{11} . Suppose quasiaffinities X, Y satisfy

$$XT = UX, \quad YU = TY.$$

If $T^n h \rightarrow 0$, then $U^n Xh = XT^n h \rightarrow 0$. Unitarity gives $Xh = 0$, and injectivity of X gives $h = 0$. Taking adjoints in $YU = TY$ gives $U^* Y^* = Y^* T^*$; since Y has dense range, Y^* is injective, and the same argument shows that $(T^*)^n h \rightarrow 0$ implies $h = 0$. Thus T is C_{11} , while Gamal's construction is not similar to a contraction.

3 Provenance and scope

This is an author-explicit literature answer: the supporting paper cites Kérchy's source as reference [K], quotes Question 18, and says the answer is negative. No new mathematical result is claimed here.

The identification settles only Question 18. The source's later questions about whether every polynomially bounded operator is quasisimilar to a contraction, the power-bounded singular-unitary problem, and invariant subspaces for full residual spectrum are not resolved by this status note.

References

- [1] L. Kérchy, *Quasianalytic polynomially bounded operators*, arXiv:1503.06691 (2015).
- [2] M. F. Gamal', *Examples of cyclic polynomially bounded operators that are not similar to contractions*, arXiv:2412.14130 (2024).